

A similarity-based framework for incipient fault detection in underground power cables

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ABSTRACT

Incipient faults are known as the faults mainly occurring on the junctions of underground power cables. Clearing quickly, these faults are difficult to detect by conventional protection equipment, leading to insulation degradation and permanent fault in the power cable over time. In this paper, employing similarity functions, a new method is put forward to distinguish the existing fault from other similar events with both high speed and accuracy. The proposed algorithm consists of a two-stages algorithm including change detection, disturbance identification, and fault detection. The proposed method performs the calculation on the voltage signal that is recorded at the sending end of the line. The cross-correlation function is employed for identifying change detection and determining disturbance time. Afterward, using sinusoidal curve fitting, the algorithm can distinguish the incipient fault from sudden load change, capacitor switching, and harmonic load. The accuracy of the proposed approach is verified by some experimental data and 6272 simulation data obtained from four types of well-known arc models, capacitor switching, and also load variations.

1. Introduction

Undetectable fault circumstances for the protection system lead to damage to the power system component and consequently, may lead to failure of the power system components. Incipient faults in underground power cables are known as one of the most significant types of faults that may not be detected by the conventional protection equipment [1]. This type of fault occurs in shorter durations [1,2], and lower peak voltages [2] in comparison with the other faults and so it is undetectable by the existing protection schemes.

The incipient faults mainly occur in parts of power cables having mechanical or insulation failure. These failures are generally developed as a result of the power cable aging, undesirable atmospheric conditions, or physical contact of the cable with other equipment [2]. It is widely accepted that the occurrence of incipient faults is accompanied by an electric arc [3]. Although electrical arcs also occur in overhead lines [4], in this paper the underground power cables are addressed. In the case of the aforementioned conditions, such faults can occur repeatedly, within time intervals. The insulation condition of the cable is damaged by each

fault occurrence, making the fault duration longer each time [1,2], and consequently, resulting in a permanent fault [5,6]. Considering that the capacitive switching, sudden system load changes [10], and also harmonic loads produce conditions similar to incipient fault incipient, the need for a powerful method, capable of detecting the incipient faults quickly and with high accuracy, is expressed.

The incipient fault is highlighted by several exclusive features, studying which can be used to identify the incipient faults and to discriminate them from other similar events:

- Have a lower peak voltage comparing to other fault types [1,2].
- Last for a very short time duration [2].
- Incept at the positive or the negative peak voltage of the sending end of the line [1].
- Usually clear at the current zero-crossing [1].
- The fault duration is reported to last from a fraction of a second to 4 cycles [1].
- The waveform of the arc during fault is closely similar to a distorted square wave [10].

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Nomenclature			
τ	Arc Time Constant	C_i	Current Constant
g	Arc conductance	n	n -th Sample Point of Voltage After Doubtful Point
u	Voltage Signal	N	Total Samples in One Period
i_{arc}	Current Signal	n_f	Number of Fault Point
P_0	Constant	i	Counter
P_1	Cooling Constant	Y_{act}	Interim Variable
U_{arc}	Constant Denoting the Mean Value of the Arc Voltage	A	Interim Variable
τ_c	Arc Time Constant In Cassie Arc Model	B	Interim Variable
g_c	Arc conductance in Cassie Arc Model	CCF	Cross-Correlation Function
τ_m	Arc Time Constant in Mayr Arc Model	HIF	High Impedance Fault
g_m	Arc conductance in Mayr Marc Model	SVM	Support Vector Machine
U_c	Constant	ST	S-Transform
		$STMV$	ST Magnitude-Time Voltage
		$STFV$	ST Frequency-Time Voltage

- The period of the fault increases after each occurrence until it leads into a permanent fault [5,6].

It should be mentioned that the detection of an incipient fault before turning into a permanent fault, helps to provide the necessary preparations for replacement of the faulty section of the cable (usually the junctions). It also helps to reduce the power supply interruption and its consequent expenses as well as collateral costs.

Several articles have been published in the areas of the incipient fault, regarding the characteristics [1–4], localization [3–9], and detection [10–18]. In [1], it is expressed that the incipient fault is associated with an arc. The authors in [2] proposed a sub-cycle incipient fault detection algorithm to prevent cable damage considering transient arcs. Utilizing wavelet transform and frequency analysis, an algorithm to detect incipient faults was proposed in [3]. Fault detection is conducted experimentally, using a special relay in [4]. In [5,6], the procedure is conducted by the extraction of the voltage wave and the current wave reactances. The fault location is detected in [7] using impedance calculation. Reference [8] proposes a mathematical approach for fault localization. The writers in [9] proposed a method for the detection of fault location, similar to [5,6], based on reactance. A Kalman filter-based method is proposed in [10], to detect the fault location in two stages, by analyzing voltages and currents at the sending end of the line. Similar to [8], based on mathematical modeling, a fault detection method is proposed by [11]. Incipient fault detection procedure is conducted using a wavelet transform in [12]. In [13], the detection of high impedance fault (HIF) is carried out by experimental data, employing time–frequency analysis. With the use of a support vector machine (SVM), and the extracted functions from wavelet transform, detection of incipient fault is conducted with high accuracy. In [16], considering three different scenarios of fault occurrence in a three-phase power cable, and using rotational current flows in the circuit, the incipient fault detection is accomplished. By upgrading a relay's algorithm, authors in [17] detect HIFs, which are of low currents. Authors in [18] provide a review of different incipient fault detection methods. In addition to providing desirable detection methods, references [15–24] proposed a better circuit modeling of incipient faults in power systems. These algorithms provided in [21–24] have mostly focused on the incipient fault location using system identification-based approaches and parameter estimation. These algorithms have provided analytical formulations with promising accuracy. In [25–27], some algorithms for the analysis of fault detection on short overhead transmission lines were proposed presented that were designed synchronized sampling, and taken into account the arcing phenomena and tower footing resistance at the fault point.

Having surveyed the publications regarding the detection and the localization of the incipient fault, the analysis of the characteristics of the incipient fault, which is similar to the momentary effect of an arc in

the system, is put forward. The references [28–40] address the analysis of standard arc models. For the sake of electric arc, the Cassie arc model, suitable for faults with high currents, was first proposed in 1939 [28]. Several years later, the Mayr electric arc model was presented [28]. Mayr proposed this model in 1943 in consideration that the arc conductivity is dependent on temperature. The Mayr model has a desirable accuracy regarding low fault currents. By series connection of the Cassie and Mayr models, the Habedank model is obtained [28]; having the superiorities of both models and therefore suitable for both high and low fault currents. The modified Mayr model, having refined the original Mayr model was also proposed in 1992 [29]. The cooling power of the Mayr model is a function of the waveform. The Schavemaker model, similar to the modified Mayr model, is derived from the original Mayr model [30]. In the Schavemaker model, the cooling power is defined as a function of its electric power and the model time constant. In [32], using the voltage-current model, a new model for the electric arc, and a formulation for the arc resistance is presented. Reference [33] analyses the detailed power-current and current–voltage characteristics of the electric arc to obtain the minimum arc incipient voltage. The authors in [34] compare the effect of different arc models for fault detection and recloser designing. Reference [35] puts forward the analysis regarding the characteristics of faults in distribution networks. Employing monitoring techniques, [36] identifies the features of the faults, occurred in power cables, and discriminates them from overhead faults. In [37], arc modeling is carried out based on their performance during switching operations in circuit breakers. In [38], a static arc model was proposed based on the analysis of laboratory arc voltage and current waveforms. In [39], a new dynamic arc model was developed based on the analysis of laboratory arc voltage and current waveforms and considered the arc length variation.

This paper presents a comprehensive framework for incipient fault detection in underground power cables and simultaneously discriminating the incipient fault from similar events. Unlike the previously published papers, the fault model in the cable is modeled based on the arc fault model. The presented framework is designed based on the similarity evaluation of voltage signal based on the well-known cross-correlation algorithm. The proposed algorithm performs the detection of the incipient fault, and its discrimination from different events in the network in two stages including 1) identifying abnormality and determining the time of disturbance occurrence, 2) discriminating the incipient fault from capacitive switching and sudden load variations, and harmonic loads. More specifically, the stages of the proposed method are as follows:

- A preliminary evaluation is conducted on the voltage at the sending end of the line to identify whether the disturbance occurs or not. Considering that the input voltage is essentially a sine wave, it can be inferred that as long as the voltage signal preserves its sinusoidal

behavior, no abnormal event has yet happened. Preliminary evaluation or so-called change detection is performed using cross-correlation of half-cycle of the recorded voltage signal.

- In the second step, once the change is detected, it is mandatory to analyze the behavior of the recorded voltage waveform to discriminate the incipient fault from similar events. Note that similar events include capacitor bank switching, sudden load variation, simultaneous starting of several electrical motors, and harmonic load change. In general, in the case of capacitor bank switching, sudden load variation, simultaneous starting of several electrical motors the voltage waveform undergoes a sudden change in the amplitude and subsequently returns to following its sinusoidal behavior by the fundamental frequency. In the case of harmonic load change, the signal is not in the standard power system sinusoidal waveform, however, the signal has a long duration continuous periodical nature. But, during an incipient fault, the voltage waveform deviates from 100% sinusoidal behavior and it has a-periodical and discontinuous behavior. The discontinuity behavior of the incipient fault stands for arc fault on/off at the fault point. To detect an incipient fault and distinguish the incipient fault from non-fault events, a one-second window of the signal including 100 half-cycles is selected to be analyzed. In the proposed detection algorithm, for each half-cycle, the recorded voltage signal is fitted by a function containing sinusoidal and DC offset components. By applying curve fitting, the so-called R-squared index is calculated. Owing to the discontinuous behavior of the incipient fault, if the R-squared index remains below a certain threshold for 1 to 25 half-cycles, the signal is identified as the incipient fault. Note in the case of harmonic load, the R-squared index remains below a certain threshold for whole 100 consecutive half-cycles since the harmonic load has a continuous behavior. Also, due to almost sinusoidal wave shape of the capacitive/inductive load changes, the R-squared index does not remain below a certain threshold for more than two consecutive half-cycles.

The contributions of the paper can be summarized as follows:

- The proposed method utilizes the direct features of the electric arc instead of not just a simple short-circuit fault. As a result, the proposed method can effectively detect incipient fault considering the minority and momentary features of this fault. Note that fault data are obtained from both simulation and experimental recorded data which effectively validates the performance of the proposed method.
- Considering the fundamental mathematics of the proposed framework, it guarantees simplicity in implementation, low complexity, speed, and accuracy comparing with training-based algorithms such as [15].
- The proposed method can distinguish other transient scenarios such as inductive load change and capacitor switching from the arc fault model in the incipient fault in the power cables. Also, while the incipient fault in the cable is repeated semi-periodically, the proposed method can discriminate the incipient fault from the harmonic load and as a result, the proposed method has robust performance in dealing with different loading and fault conditions.

In the following, Section II discusses the electrical arcs models. A description of the proposed framework is given in Section III. Performance evaluation and discussion are provided in Sections IV and V. Section VI provided some conclusions.

2. Modelling electric arcs

In order to put forward the analysis of the incipient faults more precisely, the first step is to study their inherent nature, and thereafter, employing a suitable model, analyze their characteristics. Incipient faults clear within less than a second after their incipient. Regarding the atmospheric, or environmental conditions, such faults may reoccur later

on. This process may continue repeatedly until the incipient fault is turned into a permanent fault. On the account that the atmospheric and environmental conditions vary regarding each fault event, it is not possible to provide a general model for the fault from the first occurrence of an incipient fault until the occurrence of a permanent fault. Accordingly, in this section, the modeling of this type of fault is only carried out regarding the fault duration within less than one second. Such an approach corresponds to the most realistic practice towards incipient fault simulation in the under-study system. Within the electric arc relations, u , g , and i correspond to voltage, the inverse of the nonlinear resistance, and the electric arc current, respectively. Over the years, different models have been proposed for electric arcs. In this section, some of the most conventionally employed arc models are represented.

2.1. Cassie arc model

The Cassie model was presented in 1939. This model, which is represented in Fig. 1. a, is most suitable for modeling electric arcs with high currents. In this model, described by the formula given in Table 1, τ represents the time constant and U_c is a constant.

2.2. Modified Mayr arc model

The modified Mayr model was presented in 1992. In this model, which is represented in Fig. 1.a, the cooling power depends on its corresponding electric current. The describing equation of this model is given in Table 1. The parameters τ , and C_i reflect the time constant, and the current constant, respectively, and P_0 is a constant.

2.3. The Habedank arc model

The Habedank electric arc model, which was proposed after Cassie and Mayr models, is obtained by a series connection of both mentioned models. This model, described by the equation given in Table 1, is suitable for electric arcs with both near to zero and also high currents. Fig. 1.a shows a representation of this model. The parameters g_m and g_c , reflect the conductances, and the parameters τ_m and τ_c reflect the time constants of the Mayr and Cassie arc models, respectively.

2.4. Schavemaker arc model

Similar to the modified Mayr arc model, this model was also presented by changing the Mayr arc model. In this arc model, which is described by the relation in Table 1, and represented in Fig. 1.a, the time constant, τ , is given as a parameter. Unlike previous methods, this model is dependent on the arc cooling power. This power, defined as a function of the input power, is expressed by the cooling constant, P_1 . The parameter P_0 is a constant and U_{arc} represents a constant value defined by the voltage level of the electric arc. The voltage in the sending end of the line, employing the aforementioned arc models, is illustrated in Fig. 1. b.

3. The proposed framework

The incipient fault in underground cables is highly nonlinear with random nature. Generally, an incipient fault occurs near the peak of voltage, at which the arc ignites and lasts until the arc is quenched. The occurrence and clearance of the incipient fault and consequently the fault duration have random nature. During an incipient fault, the voltage waveform deviates from 100% sinusoidal behavior and it has non-periodical and discontinuous behavior. The discontinuity behavior of the incipient fault stands for arc fault on/off at the fault point. Since the incipient fault duration is less than multiple cycles (i.e. 4 cycles), by considering a one-second window of the signal including 100 half-cycles, it is concluded that the incipient fault does not remain in all

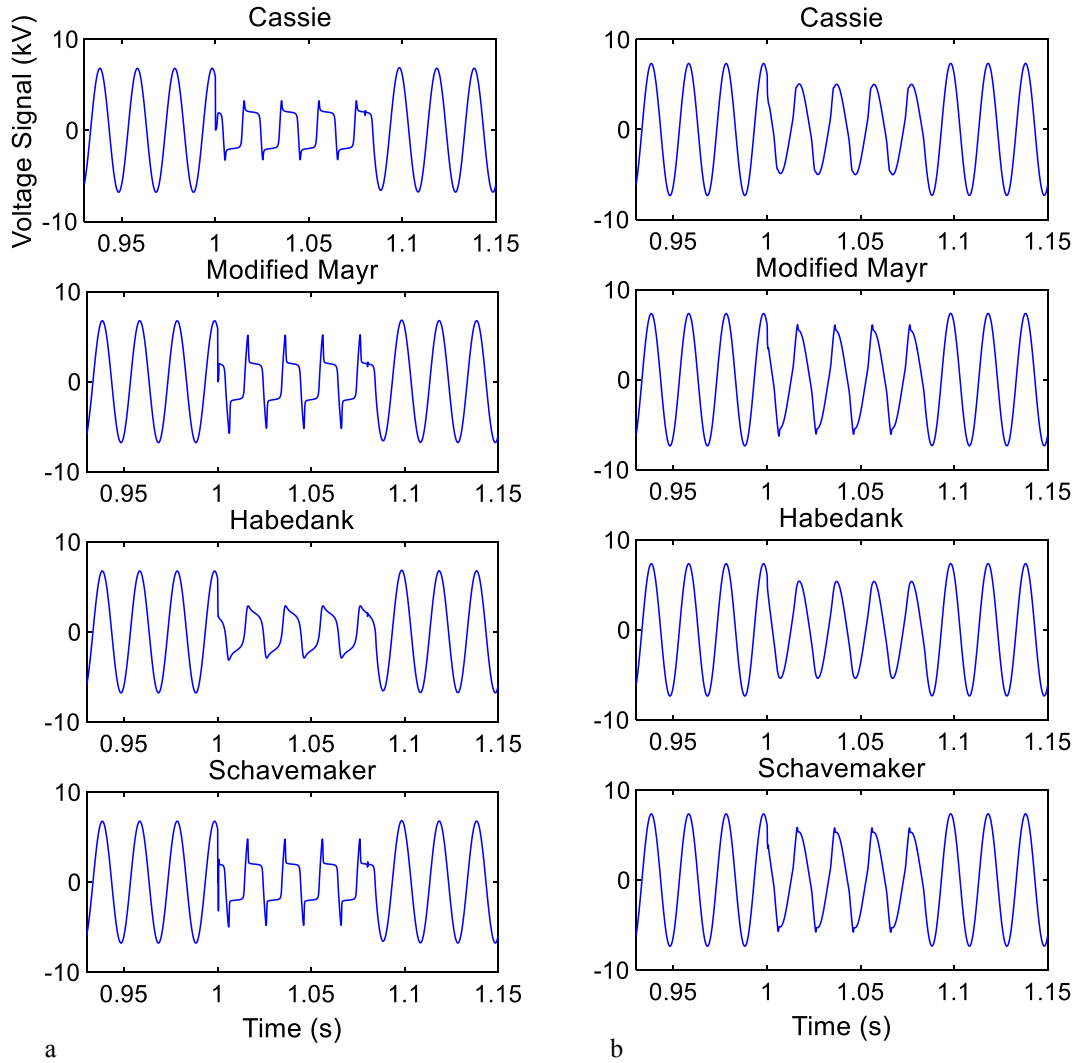


Fig. 1. a) Arc voltages at the fault incipient instance. b) Feeder voltages considering the arc in the middle of the line.

Table 1
Different arc model formula.

Arc model name	Formula	Basic parameters
Cassie	$\frac{dg}{dt} = \frac{g}{\tau} \left(\frac{u^2}{U_c^2} - 1 \right)$	$\tau = 1.2 \times 10^{-5}$
Modified mayr	$\frac{dg}{dt} = \frac{1}{\tau} \left(\frac{u i_{arc}}{P_0 + C_i i_{arc} } - 1 \right)$	$U_c = 4 \times 10^6$ $\tau = 9 \times 10^{-6}$ $C_i = 2000$
Habedank	$\frac{dg_c}{dt} = \frac{g_c}{\tau_c} \left(\frac{u^2 g^2}{U_c^2 g_c^2} - 1 \right)$ $\frac{dg_m}{dt} = \frac{g_m}{\tau_m} \left(\frac{u^2 g^2}{g_m P_0} - 1 \right)$	$P_0 = 1$ $\tau_c = 1.2 \times 10^{-4}$ $\tau_m = 1.2 \times 10^{-4}$ $U_c = 4 \times 10^6$ $P_0 = 1$
Schavemaker	$\frac{1}{dt} = \frac{1}{\tau} + \frac{g}{g_m}$ $\frac{dg}{dt} = \frac{g}{\tau} \left(\frac{u i_{arc}}{\max(U_{arc} i_{arc} , P_0 + P_1 u i_{arc})} - 1 \right)$	$\tau_c = 9 \times 10^{-6}$ $U_{arc} = 2000$ $P_1 = 0.5$ $P_0 = 0$

100 half-cycles. To reflect the randomness into account, we provided a counting-based algorithm, in which if the R-squared index remains below a certain threshold for 1 to 25 half-cycles, the signal is identified as the incipient fault. On the contrary, for harmonic load change, due to its continuous periodical behavior, the periodical behavior remains for all 100 half-cycle. As a result, the randomness of the incipient fault is the

basis of the proposed counting method for discriminating the incipient fault from the harmonic load change.

But every time that the incipient fault happens, it creates significant distortions in the voltage and current signals at the fault point. As a result of the latter distortion, the signals do not remain in the standard sinusoidal waveform. Therefore, the latter deviation from the sinusoidal wave shape can be used as the main key for detecting the incipient fault.

To be more specific, we established the proposed method based on analyzing the behavior of the recorded voltage waveform to discriminate the incipient fault from similar events. Note that similar events include capacitor bank switching, sudden load variation, simultaneous starting of several electrical motors, and harmonic load change.

The proposed algorithm is established based on the dissimilarity of the measured current signal from the standard sinusoidal waveform. The similarity is employed at two stages:

- In the first stage, the change in the signal is detected, which is conducted based on the cross-correlation similarity function. At this stage, using the cross-correlation similarity function the change is identified and some events may be filtered as the non-incipient faults (i.e. load change or capacitor switching). This is because in several load changes and capacitor switching events only a slight change in phase angle occurs and the sinusoidal waveform does not change. However, in the incipient fault and harmonic load changes, the changes in the signal are more notable and the sinusoidal waveform

changes. This is why such events are not filtered by the cross-correlation similarity function.

- The second stage is the incipient fault detection stage where the fault is recognized based on two features: (a) non-sinusoidal waveform, and (b) discontinuous nature of the incipient fault. At the second stage, to deal with feature (a), the well-known R-squared index is employed to measure the amount of dissimilarity between the measured current signal from the standard sinusoidal waveform. Knowing the R-squared index, a counting-based algorithm is provided to deal with feature (b), in which if the R-squared index remains below a certain threshold for 1 to 25 half-cycles, the signal is identified as the incipient fault. On the contrary, for the harmonic load change, due to its continuous periodical behavior, the periodical nature preserves for all 100 half-cycle. As a result, the randomness of the incipient fault is the basis of the proposed counting method for discriminating the incipient fault from the harmonic load change.

3.1. First Step: Change detection

1) Theatrical Basis of the Similarity-Based Algorithm

To mathematically illustrating the dissimilarity of the signals for the above-mentioned events at the first stage, the following justifications are provided. It is worth mentioning that in this section cross-correlation is utilized in the continuous form to drive the expressions easily. In practice, the proposed algorithm does not utilize this expression and it is implemented in discrete form.

It is assumed that the pre and post disturbances are described by $i_1(t)$ and $i_2(t)$ respectively. The continuous cross-correlation formulation is described as follows:

$$\begin{aligned} CCF &= \frac{\int_{T_w} (i_1(t) - \mu_1)(i_2(t) - \mu_2)dt}{\sigma_1 \sigma_2} \\ &= \frac{\int_{T_w} i_1(t)i_2(t)dt - \mu_2 \int_{T_w} i_1(t)dt - \mu_1 \int_{T_w} i_2(t)dt + \mu_1 \mu_2 T_w}{\sigma_1 \sigma_2} \\ &= \frac{\eta - \mu_1 \mu_2 T_w}{\sigma_1 \sigma_2} \end{aligned} \quad (1)$$

where μ and σ are mean and variance respectively and are described as follows:

$$\mu_1 = \frac{1}{T_w} \int_{T_w} i_1(t)dt \quad (2)$$

$$\mu_2 = \frac{1}{T_w} \int_{T_w} i_2(t)dt \quad (3)$$

$$\sigma_1 = \sqrt{\int_{T_w} (i_1(t) - \mu_1)^2 dt} \quad (4)$$

$$\sigma_2 = \sqrt{\int_{T_w} (i_2(t) - \mu_2)^2 dt} \quad (5)$$

In addition, η is interim variables that is described as follows:

$$\eta = \int_{T_w} i_1(t)i_2(t)dt \quad (6)$$

According to (2) to (6), μ and σ are calculated for a window length equal to T_w . In this evaluation, T_w is selected equal to $\frac{T}{2}$ where T is the fundamental period.

In the following, the CCF is calculated for two conditions:

- Condition (1) (CCF for two pure sinusoidal signals with different magnitudes and phase angles): This condition generally appears under load change and capacitor switching.

In this condition (C1), it is assumed that $i_1(t)$ and $i_2(t)$ are described as follows:

$$i_1(t) = I \sin(\omega t + \varphi) \quad (7)$$

$$i_2(t) = I' \sin(\omega t + \varphi') \quad (8)$$

where, I and I' are the magnitudes of the signals, and φ and φ' denote the phase angle of the current signals shown in (7) and (8) respectively. By substituting (7) and (8) to the expressions (2) to (6), and applying some mathematical simplifications, the essential parameters for calculating the CCF in this condition are obtained as follows:

$$\mu_1^{C1} = \frac{I}{\omega T_w} (\cos(\varphi) - \cos(\omega T_w + \varphi)) = \frac{2I \cos(\varphi)}{\pi} \quad (9)$$

$$\mu_2^{C1} = \frac{I'}{\omega T_w} (\cos(\varphi') - \cos(\omega T_w + \varphi')) = \frac{2I' \cos(\varphi')}{\pi} \quad (10)$$

$$\begin{aligned} \sigma_1^{C1} &= \sqrt{\frac{(I)^2 T_w}{2} + \frac{(I)^2}{4\omega} (\sin(2\varphi) - \sin(2\omega T_w + 2\varphi)) - T_w (\mu_1^{C1})^2} \\ &= I \sqrt{\frac{T}{4} - 2T \left(\frac{\cos(\varphi)}{\pi} \right)^2} \end{aligned} \quad (11)$$

$$\begin{aligned} \sigma_2^{C1} &= \sqrt{\frac{(I')^2 T_w}{2} + \frac{(I')^2}{4\omega} (\sin(2\varphi') - \sin(2\omega T_w + 2\varphi')) - T_w (\mu_2^{C1})^2} \\ &= I' \sqrt{\frac{T}{4} - 2T \left(\frac{\cos(\varphi')}{\pi} \right)^2} \end{aligned} \quad (12)$$

$$\begin{aligned} \eta^{C1} &= \frac{II' T_w}{2} \cos(\varphi - \varphi') + \frac{II'}{4\omega} (\sin(\varphi + \varphi') - \sin(2\omega T_w + \varphi + \varphi')) \\ &= \frac{II' T}{4} \cos(\varphi - \varphi') \end{aligned} \quad (13)$$

Using (9) to (13), the CCF for condition (1) (CCF^{C1}) can be calculated by (1).

2) Condition (2) (CCF for one pure sinusoidal signal and one harmonic distorted signal): This condition generally appears under incipient fault and harmonic load change. In this condition (C2), it is assumed that $i_1(t)$ and $i_2(t)$ are described as follows:

$$i_1(t) = I \sin(\omega t + \varphi) \quad (14)$$

$$i_2(t) = \sum_{i=1}^N I'_i \sin(i\omega t + \varphi'_i) \quad (15)$$

where, I and φ are the magnitude and phase angle of the signal expressed by (14) respectively. In addition, I'_i , φ'_i , and N denote the magnitude of i -th harmonic component, the phase angle of i -th harmonic component, and harmonic order of the current signals expressed by (15) respectively. Similarly, to the previous section, by substituting (14) and (15) to the expressions (2) to (6), and applying some mathematical simplifications, the essential parameters for calculating the CCF in this condition are obtained as follows:

$$\mu_1^{C2} = \frac{I}{\omega T_w} (\cos(\varphi) - \cos(\omega T_w + \varphi)) = \frac{2I \cos(\varphi)}{\pi} \quad (16)$$

$$\begin{aligned}\mu_2^{C2} &= \sum_{i=1}^N \frac{I'_i}{i\omega T_w} (\cos(\varphi'_i) - \cos(i\omega T_w + \varphi'_i)) \\ &= \sum_{i=1}^N \frac{2I'_i \cos(\varphi'_i)}{i\pi} \quad i = 1, 3, 5, \dots\end{aligned}\quad (17)$$

$$\begin{aligned}\sigma_1^{C2} &= \sqrt{\frac{(I')^2 T_w}{2} + \frac{(I')^2}{4\omega} (\sin(2\varphi) - \sin(2\omega T_w + 2\varphi)) - T_w (\mu_1^{C1})^2} \\ &= I \sqrt{\frac{T}{4} - 2T \left(\frac{\cos(\varphi)}{\pi} \right)^2}\end{aligned}\quad (18)$$

$$\begin{aligned}\sigma_2^{C2} &= \sqrt{\sum_{i=1}^N \frac{(I'_i)^2 T_w}{2} + \frac{(I'_i)^2}{4i\omega} (\sin(2\varphi'_i) - \sin(2i\omega T_w + 2\varphi'_i)) - T_w (\mu_2^{C2})^2} \\ &= \sqrt{\sum_{i=1}^N \frac{(I'_i)^2 T}{4} - \frac{T}{2} (\mu_2^{C2})^2}\end{aligned}\quad (19)$$

$$\begin{aligned}\eta^{C2} &= \frac{I' T_w}{2} \cos(\varphi - \varphi') + \sum_{i=2}^N \frac{I'_i}{2(i-1)\omega} (\sin(2(i-1)\omega T_w + \varphi_i - \varphi'_i) - \sin(\varphi_i - \varphi'_i)) \\ &\quad - \sum_{i=1}^N \frac{I'_i}{2(i+1)\omega} (\sin(2(i+1)\omega T_w + \varphi_i + \varphi'_i) - \sin(\varphi_i + \varphi'_i)) \\ &= \frac{I' T}{4} \cos(\varphi - \varphi')\end{aligned}\quad (20)$$

Using (16) to (20), the CCF for condition (2) (CCF^{C2}) can be calculated by (1).

After substituting the calculated parameters in the CCF expression, CCF^{C1} and CCF^{C2} can be generally expressed as follows:

$$CCF^{C1} = f(T_w, \varphi, \varphi') \quad (21)$$

$$CCF^{C2} = f(T_w, I, \varphi, I'_i, \varphi'_i) \quad (22)$$

To show the behavior of CCF for the two above-discussed conditions, the following assumptions are considered for simulating the behavior of CCF:

Condition (1): Parameters I and I' are selected randomly in the range of [1,10]. Also, φ and φ' are selected in the range of $[-\frac{\pi}{6}, \frac{\pi}{6}]$.

Condition (2): Parameter I is selected randomly in the range of [1,10]. Also, φ is selected in the range of $[-\frac{\pi}{6}, \frac{\pi}{6}]$. Furthermore, total harmonic distortion (THD) is varied between 1% and 30% considering the following states:

State 1: Only the second harmonic component is included in expression (15).

State 2: Only the third harmonic component is included in expression (15).

State 3: Only second and third harmonic components are included in expression (15).

State 4: Only second, third and fifth harmonic components are included in expression (15).

State 5: Only third, fifth, and seventh harmonic components are included in expression (15).

State 6: Only third, fifth, seventh, and ninth harmonic components are included in expression (15).

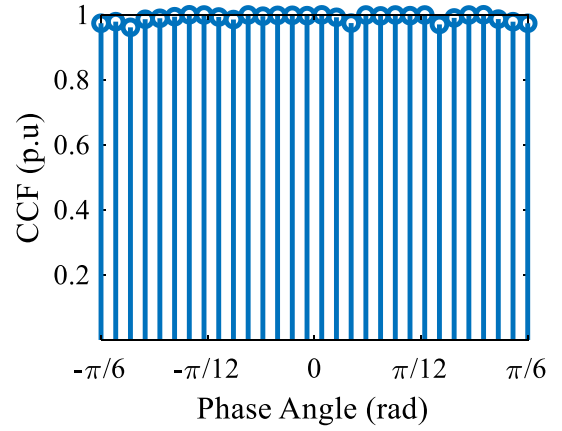


Fig. 2. the behavior of CCF^{C1} for variation given in magnitudes and phase angles.

Assuming $\omega = 2\pi f$ and $f = 50$ Hz, the behavior of CCF^{C1} and CCF^{C2} are shown in Figs. 2 and 3 respectively.

From (21) and Fig. 2, it is concluded that for conditions similar to load change and capacitor switching in which the signal before and after the change have sinusoidal waveforms, the CCF only depends on the phase angles of signals and it varies between 0.95 and 0.999. However, the expression (22) and also Fig. 3 reveal that for incipient fault circumstances, the CCF depends on both magnitude and phase angles of all harmonic components and it almost varies between 0.37 and 0.92. Note that the incipient fault and harmonic load change are separated based on the fact that in the case of harmonic load change, the signal has a long duration continuous periodical nature, but during an incipient fault, the voltage waveform has non-periodical and discontinuous behavior.

2) Implementing the similarity-based algorithm for Change Detection

In the first step, the primary evaluation is conducted on the voltage in the sending end of the line. Considering that the input voltage is essentially a sine wave, it can be inferred that as long as the voltage signal preserves the sinusoidal waveform, no abnormal event has yet happened. However, in case of the slightest deviation from sinusoidal behavior, the instance of deviation is to be reanalyzed. Accordingly, by assuming the first sample to begin the analysis from, as n , and the number of samples per cycle as N , in case the absolute voltage deviation of the n -th sample from the corresponding voltage in the previous half-cycle is within a one percent interval, no anomaly is detected and the $(n + 1)$ -th sample is put into the analysis; otherwise, more analysis is to be carried out on the n -th sample. Given the abovementioned, the following can be presumed:

$$\begin{cases} A_1 < |v(n)| < A_2 & ; \text{no events} \\ \text{otherwise} & ; \text{anomaly detection} \end{cases} \quad (23)$$

where $v(n)$ is the voltage of the n -th sample, also A_1 and A_2 are given

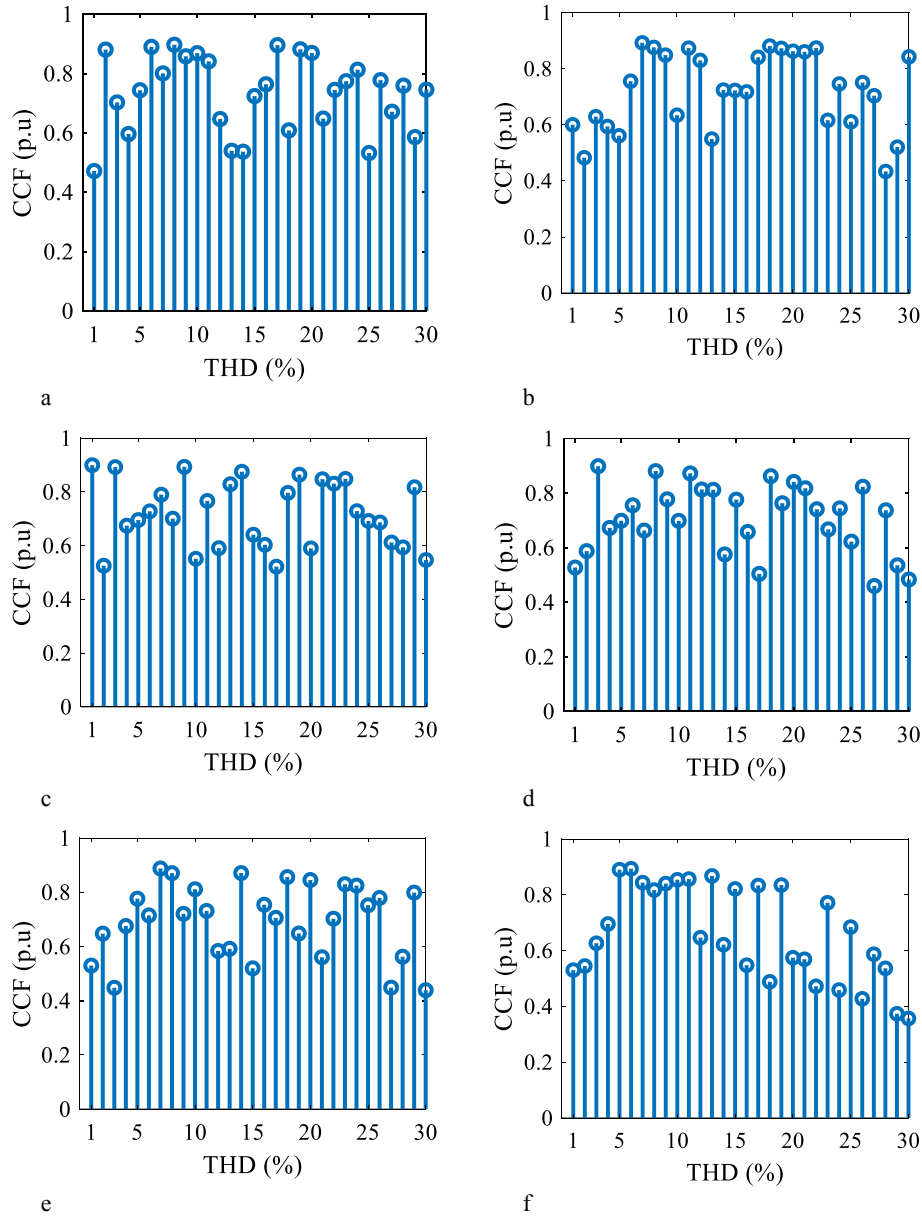


Fig. 3. the behavior of CCF^{C2} for variation given in magnitude, phase angle and THD, (a) State 1, (b) State 2, (c) State 3, (d) State 4, (e) State 5, (f) State 6.

by

$$A_1 = 0.99 \times |v(n - N/2)| \quad (24)$$

$$A_2 = 1.01 \times |v(n - N/2)| \quad (25)$$

As described previously, the incipient fault can be cleared within less than a cycle to several cycles in the system. In case $|v(n)|$ violates the condition expressed in (23), the n -th sample is put into further analysis, employing the similarity function. This paper utilizes the cross-correlation method as the similarity function which can be employed to put the similarity of two waveforms into comparison. The closer the index obtained by this function is to 1, the more similar the compared waveforms would be, and vice versa. Considering B_1 as the value of voltage samples in the sending end of the line, regarding the interval from half-cycle prior up to the point of interest, and B_2 as the value of voltage samples in the sending end of the line, regarding the interval from the point of interest to half-cycle afterward, a similarity function can be defined for B_1 and B_2 waveforms, as $CCF(|B_1|, |B_2|)$ given by:

$$B_1 = \left(v\left(n_i - \frac{N}{2}\right), v(n_i - 1) \right) \quad (26)$$

$$B_2 = \left(v(n_i), v\left(n_i + \frac{N}{2} - 1\right) \right) \quad (27)$$

$$CCF = \frac{\frac{2}{N} \sum_{n=0}^{\frac{N}{2}-1} (|B_1(n)| - |\bar{B}_1|) (|B_2(n)| - |\bar{B}_2|)}{\sqrt{\frac{2}{N} \sum_{n=0}^{\frac{N}{2}-1} (|B_1(n)| - |\bar{B}_1|)^2 \frac{2}{N} \sum_{n=0}^{\frac{N}{2}-1} (|B_2(n)| - |\bar{B}_2|)^2}} \quad (28)$$

If the cross-correlation function (CCF) reflects a value higher than 0.995, the deviation can be a result of noise or momentary inaccuracy of measurement equipment; otherwise, the point of interest is confirmed as an anomaly, and further analysis is required.

Regarding expressions (23) and (28) it should be noted that these two criteria are the supplement of each other. To be more specific, expression (1) checks the magnitude of the voltage signal at n -th sample

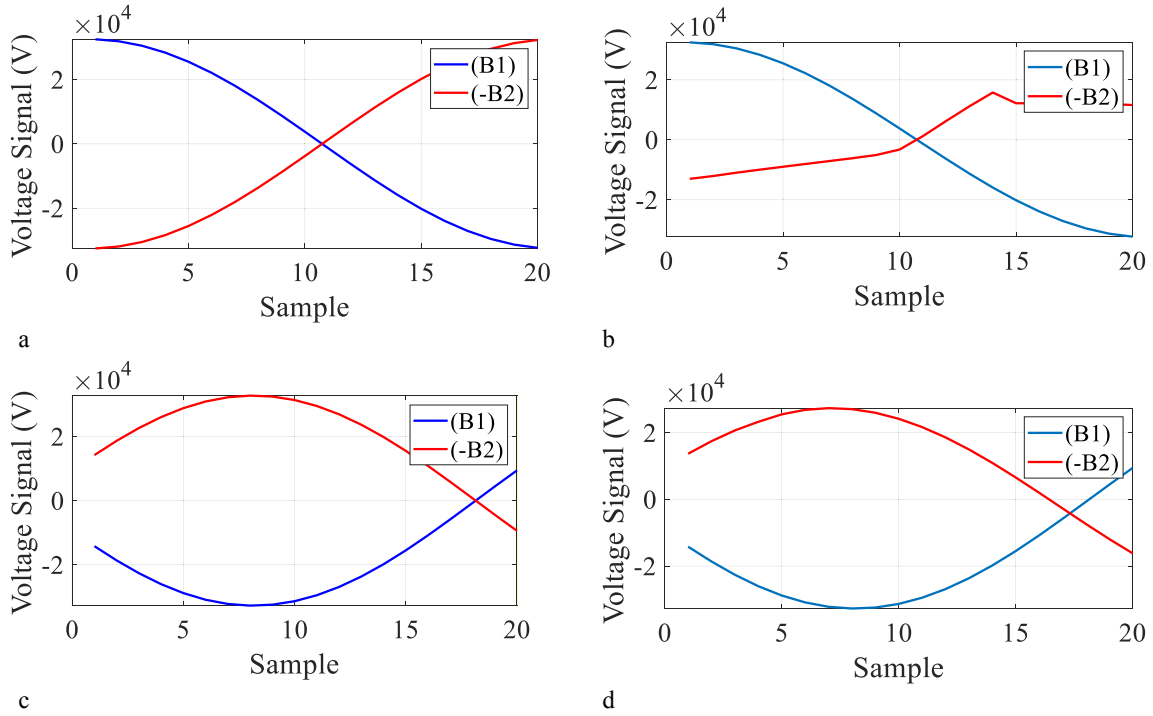


Fig. 4. a) B_1 and $(-B_2)$ before fault incipient b) B_1 and $(-B_2)$ after fault incipient, c) B_1 and $(-B_2)$ before capacitor switching, d) B_1 and $(-B_2)$ after capacitor switching.

whether if it has a change comparing with the magnitude obtained from the previous half-cycle corresponding to n -th voltage sample or not. Note that in absence of any changes, the magnitudes of the voltage signal in both half-cycles are equal. But if for one certain sample the condition of expression (1) is not satisfied, expression (6) is provided as a supplement criterion to ensure the decision making whether any changes are observed in the signal or not. While the two criteria may result in reducing speed, but the reliability and accuracy of the change detection are enhanced.

The representation of B_1 and $(-B_2)$ for the cases of incipient fault (using Cassie arc model) and capacitive switching, during the normal condition and after fault incipient is illustrated in Fig. 4. a. and b., respectively. Note that B_1 and $(-B_2)$ are selected in the illustrations of Fig. 4 to avoid full adaptation of B_1 and B_2 before disturbance occurrences. The CCF function reflects values of 1, and 0.98 for the before and after incipient fault occurrence respectively. Also, the value of CCF captures the values of 1, and 0.9996 before and after the capacitive switching correspondingly.

3.2. Second Step: Discrimination of incipient faults from similar events

According to the previous stage, once the change is detected, the behavior of the recorded voltage waveform should be analyzed to discriminate the incipient fault from similar events. Similar events include capacitor bank switching, sudden load variation, simultaneous starting of several electrical motors, and harmonic load change. From the signal wave shape point of view, the behavior of the fault and non-fault events are as follows:

- Capacitor bank switching, sudden load variation, simultaneous starting of several electrical motors: In these events, the voltage waveform undergoes a sudden change in the amplitude, and subsequently returns to following its sinusoidal behavior by the previous frequency.

- Harmonic load change: In this event, the signal is not in the standard power system sinusoidal waveform, however, the signal has a long duration continuous periodical nature.
- Incipient fault: In this event, the voltage waveform significantly deviates from sinusoidal behavior and it has a-periodical and discontinuous behavior. Note that the latter discontinuity behavior of the incipient fault reflects the arc fault on/off behavior at the fault point.

To detect an incipient fault and distinguish the incipient fault from non-fault events, a one-second window of the signal including 100 half-cycles is selected to be analyzed. In the proposed detection algorithm, for each half-cycle, the detection algorithm operates based on the waveform-related index and the continuity behavior of the recorded voltage signal. In the following, the waveform-related index is described.

As previously described, the proximity of the voltage signal recorded in the sending end of the line to a sine wave, after event occurrence is put into analysis. In this stage, a sine wave is fitted to the voltage signal in the sending end of the line, within a half-cycle after the event incipient. The fitted sine wave is described by the relation in (29), as:

$$y = f(x) = a_0 + a_1 \cos(\omega x) + b_1 \sin(\omega x) \quad (29)$$

The waveform fitted to the recorded voltage in the sending end, after fault incipient, and regarding other similar events is represented in Fig. 5. The parameter estimation is conducted based on the MATLAB curve-fitting toolbox so-called CFTOOL. The CFTOOL employs the Levenberg-Marquardt algorithm to solve nonlinear least-squares problems [41]. After fitting a sinusoidal waveform to the recorded voltage samples and calculating the a_0 , a_1 , and b_1 coefficients, the R-squared value regarding the estimation made is calculated. The R-squared value is calculated as follows:

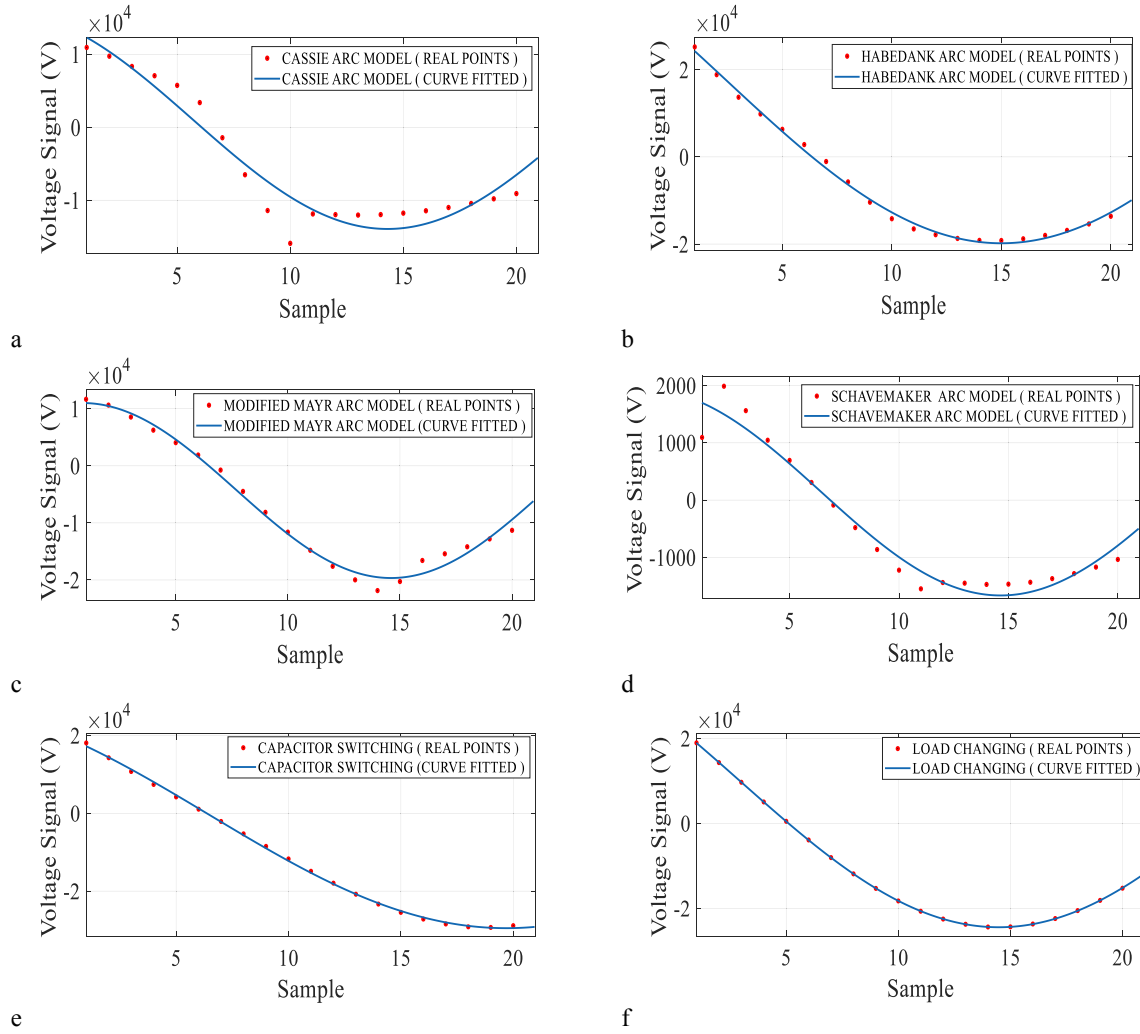


Fig. 5. Fitting a sine wave to the voltage signal recorded within half-cycle from event incipient; incipient fault and other events. (a) Cassie Arc Model, (b) Schavemaker Arc Model, (c) Modified Mayr Arc Model, (d) Habedank Arc Model, (e) Capacitor Switching, and (f) Load Changing.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (30)$$

where, y_i and $\hat{y}_i$ are the i -th actual and estimated values of the voltage signal. Also, $\bar{y}$ is the mean value of actual data that is obtained for N samples as follows:

$$\bar{y} = \frac{\sum_{i=1}^N y_i}{N} \quad (31)$$

This value is in correspondence with the maximum accuracy of the fit, considering the recorded values. The closer this value is obtained to 1, the more accurately the sinusoidal estimation of the recorded data has been made, and accordingly, the closer to 0 should the value be obtained, the lower the sinusoidal estimation accuracy would be. In this stage, the aim is to evaluate the similarity of the recorded signal to a sine wave. Therefore, the R-squared index obtained for the voltage in the sending end of the line is assessed. Having calculated an R-squared greater than 0.999 represents a high similarity of the recorded waveform to a sine wave. The conditions reflecting such R-squared values correspond to sudden system load changes and capacitive switching events. As a result of their non-harmonically sinusoidal nature, such loads follow their previous sinusoidal behavior, after event occurrence. On the

other hand, in case the R-squared index is calculated to be less than the above mentioned, it is inferred that the voltage in the sending end of the line momentarily deviates from following its sinusoidal behavior, and harmonics are generated. The data regarding the estimated values for a_0 , a_1 , and b_1 coefficients, and the parameters of ω and R-squared for the waveforms represented in Fig. 5 are given in Table 2.

To have better described this method, the flowchart diagram of the

Table 2

The estimated parameters for the fittings, represented in Fig. 5.

General model: $f(x) = a_0 + a_1 \cos(\omega x) + b_1 \sin(\omega x)$					
File name	R-squared	a_0	a_1	b_1	W
Cassie Arc Model	0.9319	969.4	1.29E + 04	-7426	0.1832
Schavemaker Arc Model	0.9580	137.3	1703	-585.8	0.1919
Modified Mayr Arc Model	0.9886	-4328	1.50E + 04	2952	0.2284
Habedank Arc Model	0.9861	1.75E + 04	1.13E + 04	-3.5E + 04	0.1253
Capacitor Switching	0.9992	-422.2	2.03E + 04	-2.1E + 04	0.1189
Load Changing	0.9999	9762	1.36E + 04	-3.1E + 04	0.1366

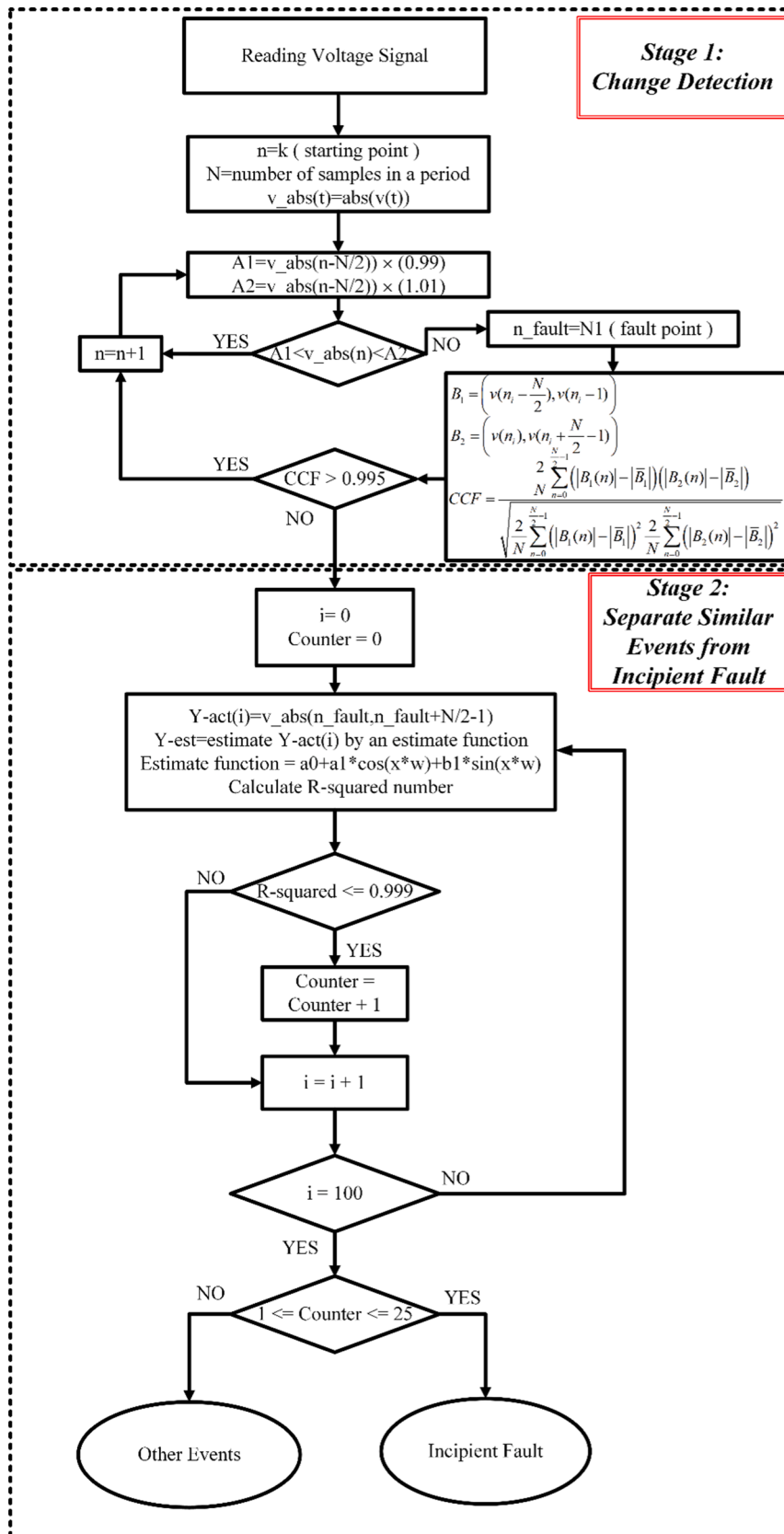


Fig. 6. The flowchart of the proposed method.

proposed method is depicted in Fig. 6. According to Fig. 6, after applying change detection, for 100 half-cycle equal to 1 s, if the counter remains between 1 and 25, the incipient fault is identified.

3.3. Discussion on the threshold setting

The proposed method utilizes three thresholds for the discrimination between incipient fault and load change in the underground cables. The thresholds are dependent on the waveform characteristics and the similarity to the sinusoidal waveform. As mentioned in the manuscript, the threshold value is selected based on plenty of simulations in different scenarios. To obtain threshold, an algorithm named “Otsu thresholding method” was utilized which is a well-known and reliable method, employed in different engineering fields [42–46].

In the following, the implementation of “Otsu thresholding method” in the proposed method is described in more detail.

- 1) In the first step, a Probability Function Density (PDF) is assigned to the desired parameter for different conditions such as incipient fault and load changes scenarios (in the proposed method, PDFs should be assigned for CCF and R-squared).
- 2) In the second step, a normal function-based curve should be fitted for each case.
- 3) In the third step, the intersection point of the PDF curves regarding the normal and fault cases is selected as the threshold value.

The thresholds are obtained from simulated samples under various conditions including:

- 1) Applying arc fault at different fault location (from 1 km to 9 km from the voltage recorder)
- 2) Applying different arc fault at different time instance (positive polarity voltage peak, negative polarity voltage peak, zero crossings of the voltage signal, random instance of voltage signal)
- 3) Applying different arc fault with different time constant during fault
- 4) Applying different load changes at a different location from voltage metering
- 5) Applying multiple motor starting to simulate entrance of parallel loads at the end of the cable
- 6) Applying different capacitive, inductive load changes and motor starting at different time instance (positive polarity voltage peak, negative polarity voltage peak, zero crossings of the voltage signal, random instance of voltage signal)

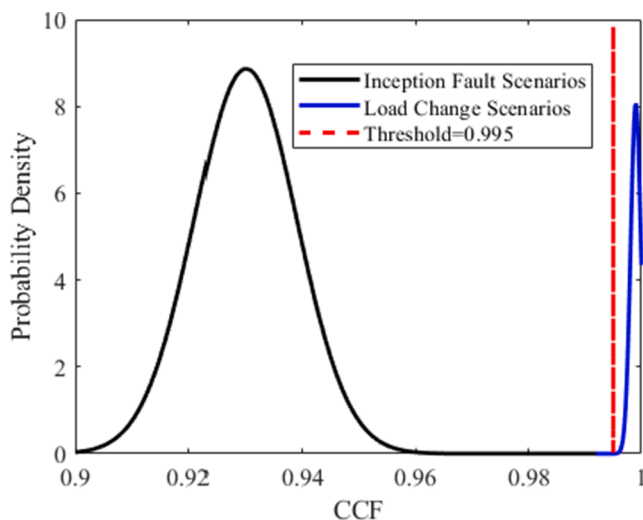


Fig. 7. PDFs for the index *CCF* to detect change detection and the selected threshold.

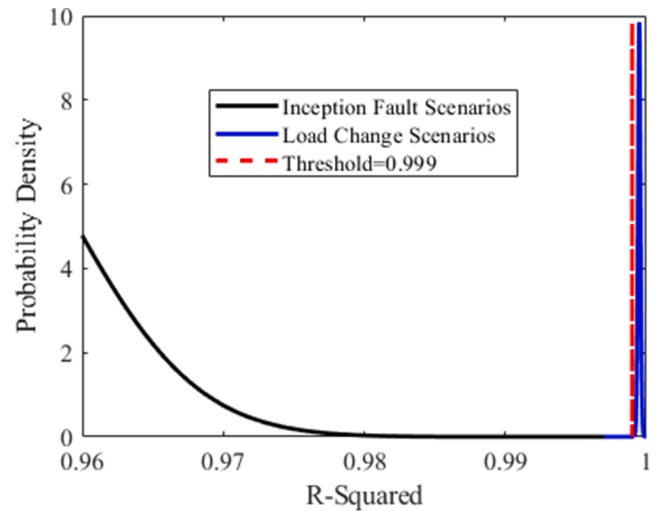


Fig. 8. PDFs for the index *R-Squared* to discriminate capacitive/inductive load change from incipient fault and the selected threshold.

- 7) Considering the different level of random noise with the signal to noise ratio between 40 dB and 60 dB

As one can see in Figs. 7 and 8, in our investigation, the PDFs of stages 1 and 2 for incipient fault scenarios and non-fault scenarios have an intersection. Therefore, the Otsu thresholding method can be used to show the obtained threshold is valid under different circumstances.

4. Simulation and experimental results

4.1. The simulated system and results

One of the most important stages of each simulation is its data. A part of this data is acquired experimentally, and the other part is obtained by simulation of conditions similar to real scenarios. Regarding the latter, the more realistically the simulation of the condition is carried out, the more valuable the obtained data would be, and accordingly, the results obtained from the analysis of these data would be more reliable. The under-study system in this paper is similar to a real scenario from the existing experimental data having a 10 kV line-to-line voltage in the input branch. This system contains a three-phase underground cable, with a horizontal structure. The incipient fault occurs along this cable and within one phase. Considering three different scenarios, most of the possible conditions are considered and further analyzed. The input voltage is logged by a data logger. The simulation step time is considered 50 μ s, however, by further analysis, and having a reasonable error with fewer data, only 4000 samples from the 40000-sample data recorded within 2 s is utilized (actually, the evaluation is conducted by 500 μ s simulation step time). The one-line diagram of the three-phase under-study system and the details from the cable used are shown in Fig. 9.

The specifications for the under-study system are as follows:

- $V = 10$ kV
- Frequency = 50 Hz
- Load = 0.022 (H) + 9 (Ω) (assumed 500A current in consumer side)
- Length of line = 10 km
- Cable size = 1x300 / RM 25
- Time simulation = 2 sec
- Fault duration = 4 cycle or 0.08 sec

The properties of the under-study cable are as follows:

- Standard IEC6052
- Layers = CU/SC/XLPE/SC/SCT/CWS/PVC

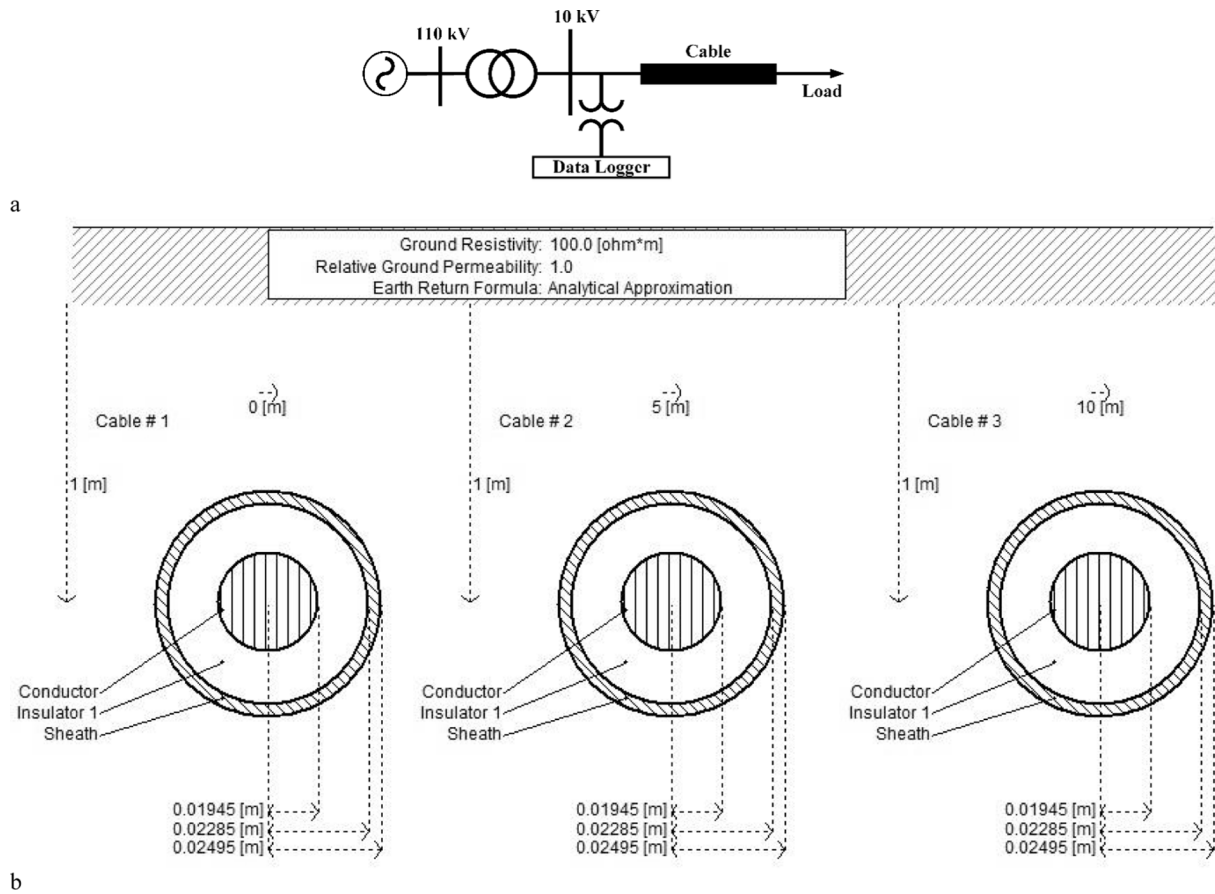


Fig. 9. a) The one-line diagram for the feeder branch embedding the incipient fault. b) The three-phase cable simulated in PSCAD.

- (Because of software limitation for simulation, just 3 layers of it simulated that contain Core, Insulation, Sheath)
- Cable size = 1x300 / RM 25 (25 stranded cable)
- The thickness of the insulation layer = 3.4 mm
- The thickness of the sheath layer = 2.1 mm
- The core diameter = 38.9 mm
- The weight of cable = 3644 kg/km
- The distance between cables = 0.5 m
- The depth of burial = 1 m

The recorded voltage and currents are logged using a data logger, which has a sampling time 50 μ s and it can be set to log voltages and currents of the system in an abnormal condition. The selection of the voltage and current sensors and their locations have an essential role in the monitoring system. Some of the important factors for the selection of the sensors are as follows:

- Suitable frequency response
- Cost-effectiveness
- Size
- Noise and interference immunity

Table 3

The results of the proposed approach in the detection of incipient faults by different arc models and similar events.

Tested Model	Number of Cases	Accuracy of Detection
Cassie	1320	100%
Modified Mayr	1305	99%
Habedank	1320	98%
Schavemaker	1320	99%
capacitive switching	347	98%
sudden load change	660	100%

(v) PT accuracy class

However, it should be noted that the proposed method does not require any specific measurement device and it can be implemented using already installed PTs in the system.

As mentioned before, the incipient fault locations in underground cables are usually in the junction of the cables. On this ground, each case of simulation is conducted through the utilization of each arc model on a point along the cable length to the ground, and the voltage is logged on the sending end. Such a process is carried out regarding each model, considering different scenarios, and a total of 6272 cases of data are generated. In addition to the data regarding the simulation of incipient faults, utilizing different arc models, capacitive switching events, and sudden load changes in the system are also simulated using the presented circuit, by considering three different scenarios, described in the following. The results of the aforementioned are tabulated in Table 3.

Within the present scenarios considered for incipient faults using each arc model, three scenarios, being the most comprehensive, are employed to acquire data from the system:

- Implementing one of the arc models within different distances along the under-study cable branch (within 1 km to 9 km from the sending end).
- Change in time constants of the arc model considered, and consequently change in waveform of the electrical arc during a fault.
- Change in the duration in which the arc is applied to the circuit and consequently the incipient of arcing fault within different intervals from positive to negative peak of the main input voltage.

Accordingly, three other scenarios are considered for the sake of data collection regarding similar events with behaviors in resemblance of incipient fault (capacitive switching events and sudden load variations),

Table 4

The first step result of the proposed method by using the Cassie arc model as an incipient fault in the 6 km from monitoring point.

Model Name	$\tau(\text{sec})$	Time(s)					
		1	1.002	1.004	1.006	1.008	1.01
Cassie Arc Model	5×10^{-6}	0.9584	0.9616	0.9608	0.9572	0.9438	0.9018
	9×10^{-6}	0.9547	0.9561	0.9368	0.9618	0.9478	0.9086
	13×10^{-6}	0.9687	0.9582	0.9661	0.9602	0.9545	0.9403
Modified Mayr Arc Model	50×10^{-7}	0.9519	0.9482	0.9638	0.9434	0.9208	0.9426
	58×10^{-7}	0.9381	0.964	0.9582	0.9398	0.9355	0.9515
	66×10^{-7}	0.9354	0.9573	0.9580	0.9495	0.9475	0.9402
Hadedank Arc Model	5×10^{-6}	0.956	0.9635	0.9659	0.9581	0.9537	0.9483
	9×10^{-6}	0.9593	0.9619	0.9673	0.9619	0.9541	0.9516
	13×10^{-6}	0.9416	0.9665	0.9626	0.9292	0.9651	0.9553
SchavemakerArc Model	5×10^{-6}	0.9339	0.9603	0.9630	0.9483	0.9483	0.9624
	6×10^{-6}	0.9594	0.9436	0.9638	0.9513	0.9244	0.9695
	7×10^{-6}	0.9214	0.9698	0.9537	0.9484	0.9056	0.9655

Table 5Capacitor Switching ($L = 0.022\text{H}$) and load changes in the 4 km from the monitoring point.

Event Name	Changing Parameters		Time(s)					
			1.000	1.002	1.004	1.006	1.008	1.010
Capacitor Switching	Capacitor (F)	450	0.9991	0.9993	0.9993	0.9994	0.9997	0.9983
		476	0.9988	0.9996	1	0.9996	0.9991	0.9988
		502	0.9981	0.9999	0.9997	1	0.9988	0.9981
Load Changing	Load = $R + jL\omega$	$17 + j0.001\omega$	0.9991	0.9992	0.9999	0.9996	0.999	0.9999
		$17 + j0.008\omega$	0.9994	0.9991	0.9994	0.9998	1	0.9993

as follows:

- Entering capacitor banks and variation of electrical loads in the system within different time intervals (from 1 to 1.02 s)
- Utilizing different loadings in the consumer end of the line, and as a result, different capacitor values for reactive power compensation.
- Sudden load variations in the receiving end of the power cable, utilizing different load values and applying the load changes by 10 to 0.1 times the rated power.

As mentioned, 6272 scenarios for different arc models and disturbance incipient time are generated according to the above-mentioned conditions and then are applied to the proposed method. The proposed method can provide detection very fast and accurately and also provide discrimination from other events. In the case of load switching both inductive and capacitive, the proposed method successfully identified the load changes scenarios from fault scenarios. Overall, the accuracy of the proposed approach is about 99%.

The numerical results of CCF described in Tables 4 and 5 are only a small section of the test results regarding the first stage of the proposed method, and well represents the high capability of this method in the detection of incipient faults.

The proposed method is a similarity-based approach. The dissimilarity of arc fault models to sinusoidal waveform as the normal condition is the main key to find the incipient fault. There are some issues such as inaccuracy in the voltage meter, data logger, sampling rate, and even analog-to-digital and digital-to-analog processes which can induce inaccuracy in the recorded signal. It is obvious if the voltage signal is recorded more accurately, the correct incipient fault detection rate will be enhanced. Moreover, there are three stages of classification in the proposed algorithm. In each stage, there is one threshold and the threshold-based classification may suffer from some misclassification during the process. Such a misclassification may be imposed by a higher level of noise or some calculation errors that any methods may be encountered with.

4.2. Performance evaluation under harmonic load

To show the performance of the proposed method under harmonic load conditions, three recorded voltage signals of the cable that feeds an electric arc furnace (EAF) are provided in Fig. 10. The EAF is known as the largest nonlinear harmonic loads which contain very reach even, odd and inter-harmonic contents that its harmonic components have time-variant behavior. According to Fig. 10.a, c, and e, the voltage signals data measured at the cable sending end are provided for a one-second duration. As one can see in Fig. 10.b, d, and f, the R-squared index for three scenarios has continuous behavior for 1 s, and the R-squared index fall below the threshold for 1 s (100 half-cycle). As a result, the proposed algorithm identifies the signals as the harmonic load.

4.3. Performance evaluation with experimental data

In order to put forward the analysis of the proposed method as accurately as possible, after simulation data, the presented method is analyzed using the experimental data. In the experimental under-study system, there had been disturbances observed in electronic power loads and also UPSs. The real-life data are adopted from [10]. More specifically, In Fig. 11.a a single line diagram of a part of the network, in which the field data have been measured, is presented. In this figure, some information on the system and the considered feeders are presented. The measured data are recorded from F2, which has a 6 km underground cable (type XLPE).

The test system has the following specifications:

- 110 kV grid: $V_n = 110 \text{ kV}$; Short Circuit Positive Sequence Impedance: $0.7 + j6.5 \Omega$.
- 110/10 kV transformer (YY): Nominal Rated Apparent Power: 50 MVA; Impedance Voltage = 12%.
- F1, F2, and F3 (Overhead lines): AL/ST, 70/12, 24 km in total.
- F2 (Underground cable): N2XS2Y, 3x1x185, 6 km in total.
- F4 (Cable): NKBA 3x70, 3 km.
- Capacitor bank: 1 MVAR, 3MVAR, 4MVAR.

For the sake of a more thorough analysis, both the voltages and the

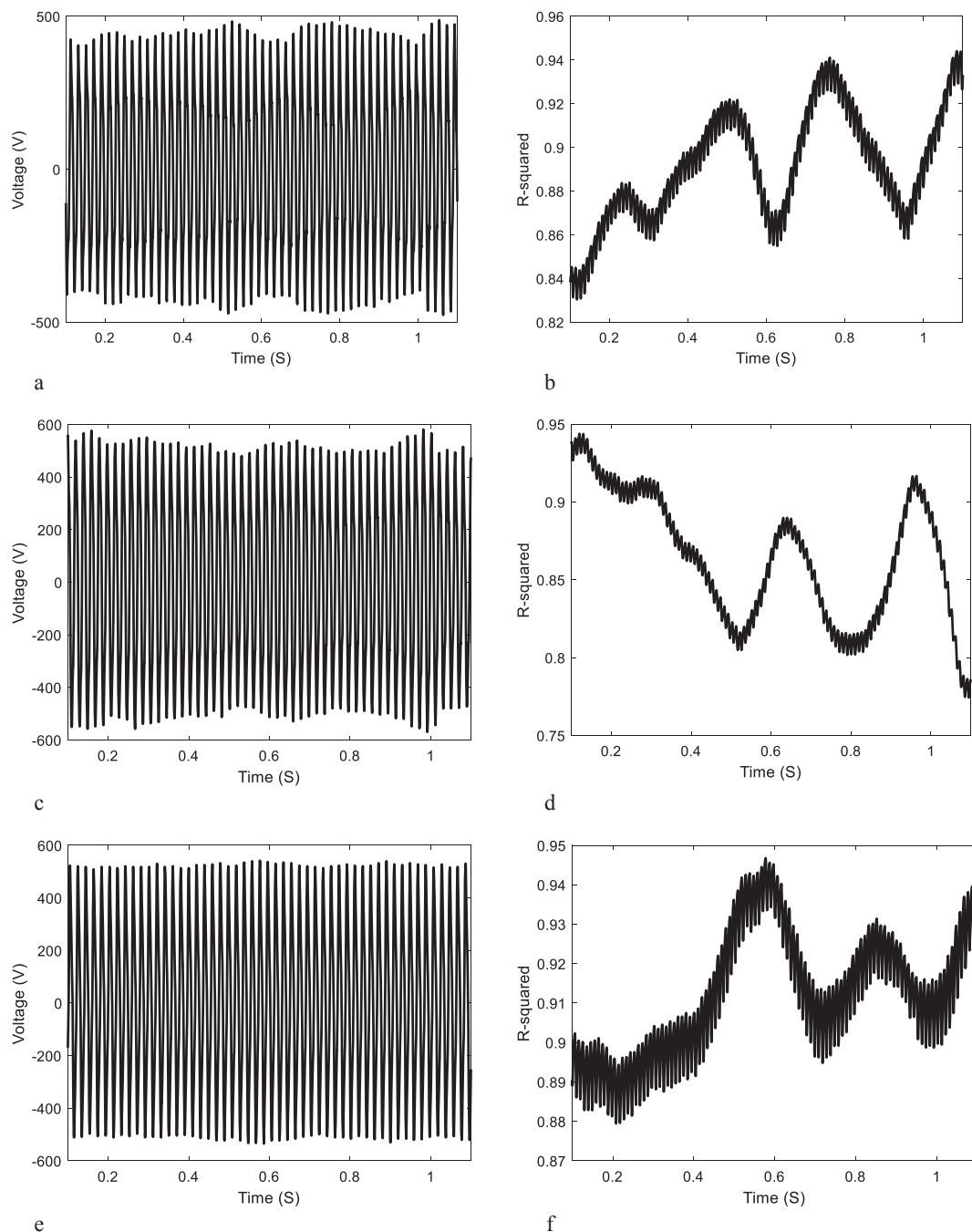


Fig. 10. Performance evaluation of the proposed method for harmonic load condition using field data measurement (a), first voltage signal, (b) R-squared index for the first voltage signal, (c), second voltage signal, (d) R-squared index for the second voltage signal, (e), third voltage signal, (f) R-squared index for the third voltage signal.

currents in the sending end of the cable were logged. Given that the breakers had not operated, and also that the voltage waveform had represented behaviors similar to that of incipient faults (the logged voltage contained waveforms similar to square waves with high amplitudes), further analysis had carried out and considering that the incipient faults commonly occur in the junction point of the cables, backup teams were sent to the field. Note that the proposed incipient fault detection was only performed on the voltage signal. The team had explored the junction and had observed an audible noise from the junction section of the cable. Thereafter the cable had been de-energized and the junction had been analyzed by the team. Having observed insulation degradation in the junction area, the occurrence of incipient

fault had been verified [10]. The data acquired from the experimental case consists of 3 waveforms. The proposed approach can provide fast and accurate detection of incipient faults in this system, likewise in the simulation cases. The detection stages regarding the experimental data are depicted in Fig. 12 (the fault incipient instance is considered second 1, likewise for the simulation data had been considered).

As illustrated in Fig. 12.a, the primary analysis of sample 497 shows voltage deviation from the corresponding voltage from one half-cycle prior. Therefore, the parameters B_1 and B_2 , are calculated, and thus the CCF is obtained to be 0.32, raising suspicion of fault incipient on this sample. Thus, a sinewave is fitted to the half-cycle interval from this sample, by the R-squared of 0.528. Having calculated the R-squared, the

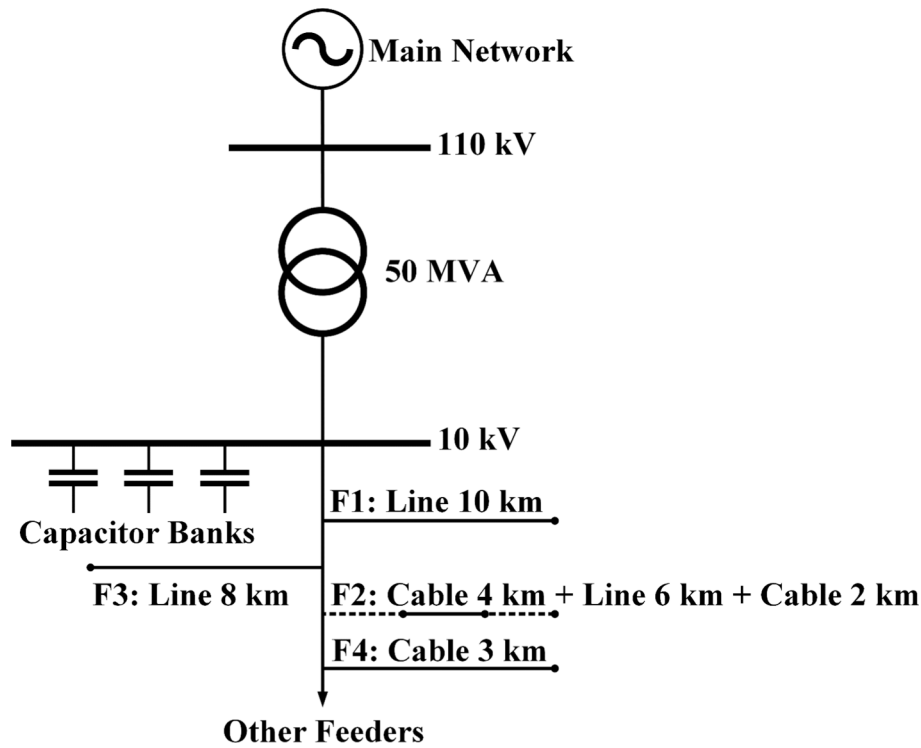


Fig. 11. a single line diagram of a part of the network for the field data measurement.

iteration counter stops at 8, and the incipient fault is detected in the system.

The employed detection criteria regarding the experimental data are tabulated in Table 6.

5. Comparison with the State-of-The-Art

The comparison with the previously suggested algorithms for incipient fault identification is provided in the following. Such comparison is provided to represent the capabilities of the proposed approach in comparison to other similar methods. The selected methods from state-of-the-art are described in the following.

5.1. First method: (Kalman Filter)

An incipient fault detection method is provided in [10] by which the voltages and currents are measured in the sending end of the line, and are estimated with the use of a Kalman filter. The variance of the differences between the measured voltages and their corresponding estimation are provided in C_m , and the differences between the measured and the estimated currents are provided in R_n . Now, by comparing these values to the threshold value, obtained from various tests, the final decision is taken. In the first step, the comparison to the threshold value is conducted for current, and in case of confirmation, the voltage is then analyzed. If both stages confirm, the event is verified as an incipient fault. It should be mentioned that when the values are obtained smaller than the threshold, the occurred event would not be an incipient fault. Given the two-stage calculation, this method is more powerful in comparison to other methods. Even though, this method is not as powerful as the proposed method.

5.2. Second method: (Combination of S-Transform and SVM)

Utilizing support vector machine (SVM), and the s-transform matrix, an incipient fault detection method is introduced in [15]. In this method, using the s-transform (ST) matrix, indexes ST of magnitude-time voltage

(STMV), and ST of frequency-time voltage (STFV), as fully described in the main paper, are calculated in each phase. Afterward, these indexes are compared to threshold values regarding each index, the final decision regarding the incipient fault occurrence is taken. Moreover, this paper provides other indexes such as STMV and STFV, all of which are obtained through trial and error, and other tests. This method employs the SVM for verification, and by clustering the data using the kernel function $e^{-\frac{\|x-y\|^2}{2\sigma^2}}$ within a multidimensional space, takes the final decision of incipient fault detection. Although the method of this paper represents high accuracy, as of being a multistage method, however, most of the indexes provided in this method are obtained by trial and error. Moreover, even though the provided laboratory data confirm this method, a slight change in the real condition can violate all the provided indexes which are obtained from the provided data.

The results regarding the analysis of the provided 6272 data, using the above-mentioned methods, and also the proposed approach is tabulated in Table 7. As can be seen, the proposed method represents high accuracy, on the account that it directly utilizes the features of the incipient faults.

This method can detect the incipient fault incipient within only a half-cycle from fault incipient (0.01 s), which shows the superiority of this method over other similar methods.

6. Conclusion

Due to a short duration, identifying the incipient faults in the underground cables has become a challenging circumstance for the protection relays. In this paper, the incipient fault detection was carried out with the help of similarity functions, using the voltage recorded at the sending end of the line. In the first step, the disturbance incipient time was detected with high accuracy. Afterward, by fitting a sinusoidal waveform to the data logged after the event incipient, the similarity of the recorded data to capacitive switching and sudden load variation waveforms was analyzed. By analyzing the repetition of the absolute of the recorded waveform in each half-cycle, the final decision regarding the incipient fault occurrence was taken. Note that the incipient faults

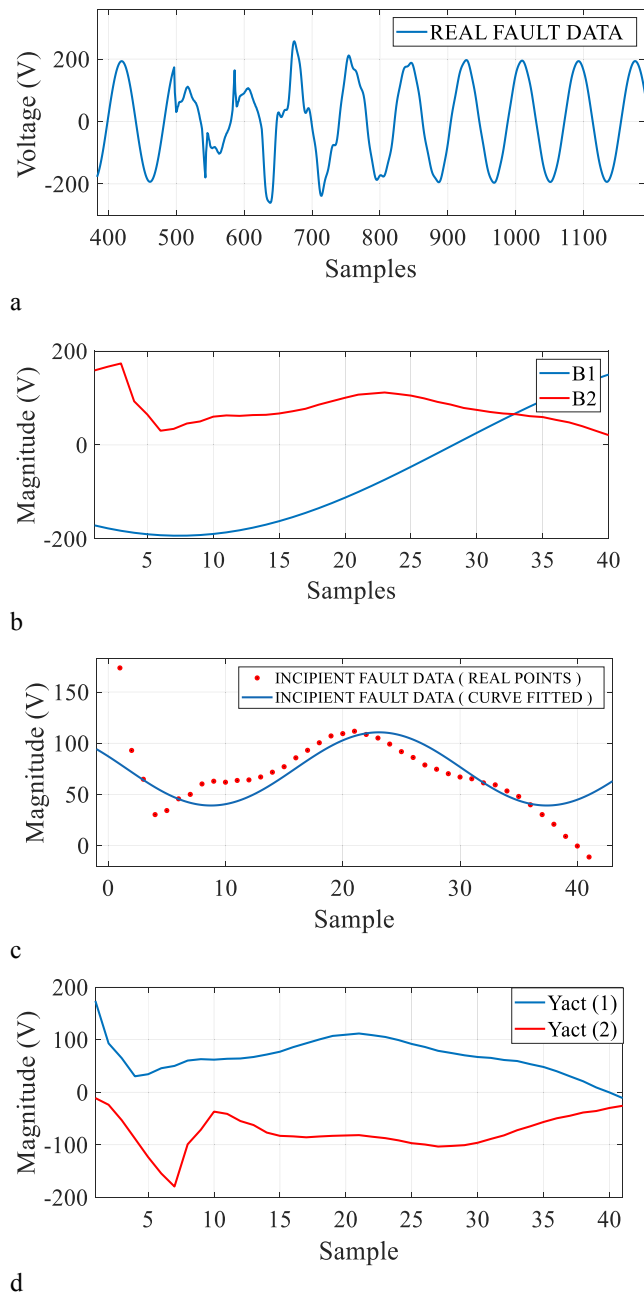


Fig. 12. a) Fault signal, b) B1 and B2 for experimental data of incipient fault, c) Half-cycle fitting of the sine wave from event incipient sample, c) $Y_{act}(1)$ and $Y_{act}(2)$ for experimental incipient fault case.

Table 6

The parameters of the proposed method for detection of the 3 real incipient fault cases.

Real data No. Detection steps	1	2	3
CCF(1)	0.32	0.42	0.80
R-squared	0.528	0.671	0.821

are cleared within several cycles (i.e. four or five cycles). As a result, if the waveform of interest was repeated less than a certain threshold, the occurred event was identified as an incipient fault, and otherwise a harmonic load. Based on the several simulations and experimentally recorded data, it has been concluded that:

(i) The proposed method is capable of discriminating incipient faults

Table 7

Comparison between the proposed approach, and the previous methods.

Method	Description	Detection Accuracy
Reference [10]	Kalman filter	92.1%
Reference [15]	SVM and S-Transform	90.1%
Proposed method	–	99.2%

from other similar system events

(ii) The proposed method has a low computational burden.

(iii) The proposed method has a notable accuracy in the detection of incipient faults and requires very few samples for the detection (only a half-cycle).

(iv) Since the proposed method is independent of incipient fault waveform, this approach is capable of fault detection with high accuracy given change in the atmospheric conditions and the arc extinguishment.

(v) Comparison with the state-of-the-art, the proposed method provides more accurate results in different fault conditions.

The limitations and future works of this paper are summarized as follows:

- Future work would be dedicated to the implementation cost of the measurement units as well as the required data logger and processing system are the main challenges in practice, which should be investigated to be reduced.
- As the fault location becomes far from the measurement point, the attenuations of the incipient fault characteristics increases. Future research can be dedicated to employing the approximation of the fault point voltage signal in incipient fault detection. Since the incipient fault characteristics at the fault point are less attenuated in the approximated fault point data comparing with the sending end of the power cable, the reliability of the incipient fault detection will be expected to enhance.
- The proposed method is implemented for a single feeder. Future work can be dedicated to allocating measurement units and implementing an algorithm that utilizes the measurement units for incipient fault detection.
- Due to the nonlinear behavior of HIF, it seems the proposed similarity-based algorithm can be modified and applied for the HIF detection.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Sidhu Tarlochan S, Zhihan Xu. Detection of incipient faults in distribution underground cables. *IEEE Trans Power Delivery* 2010;25(3):1363–71.
- [2] Kojovic Ljubomir A, Williams CW. In: *In 2000 Power Engineering Society Summer Meeting (Cat); 2000. p. 1175–80.*
- [3] Benner, C., K. B. Purry, and B. D. Russell. "Distribution fault anticipator." EPRI, Palo Alto, CA 1001879 (2001).
- [4] Terzija VV, Radojevic ZM. Numerical algorithm for adaptive autoreclosure and protection of medium-voltage overhead lines. *IEEE Trans Power Delivery* 2004;19(2):554–9.
- [5] Kim Charles J, Bialek Thomas O. Sub-cycle ground fault location—Formulation and preliminary results. In: *In 2011 IEEE/PES Power Systems Conference and Exposition; 2011. p. 1–8.*
- [6] Jia Ke, Thomas David, Sumner Mark. A new single-ended fault-location scheme for utilization in an integrated power system. *IEEE Trans Power Delivery* 2012;28(1):38–46.
- [7] Filomena André D, Resener Mariana, Salim Rodrigo H, Bretas Arturo S. Fault location for underground distribution feeders: An extended impedance-based formulation with capacitive current compensation. *Int J Electr Power Energy Syst* 2009;31(9):489–96.
- [8] Torres V, Guardado JL, Ruiz HF, Maximov S. Modeling and detection of high impedance faults. *Int J Electr Power Energy Syst* 2014;61:163–72.

- [9] Kulkarni Saurabh, Santoso Surya, Short Thomas A. Incipient fault location algorithm for underground cables. *IEEE Trans Smart Grid* 2014;5(3):1165–74.
- [10] Ghanbari Teymoo. Kalman filter based incipient fault detection method for underground cables. *IET Gener Transm Distrib* 2015;9(14):1988–97.
- [11] Zhang Wenhai, Jing Yindi, Xiao Xianrong. Model-based general arcing fault detection in medium-voltage distribution lines. *IEEE Trans Power Delivery* 2016;31(5):2231–41.
- [12] Baqui Ibrahim, Zamora Inmaculada, Mazón Javier, Buigues Garikoitz. High impedance fault detection methodology using wavelet transform and artificial neural networks. *Electr Power Syst Res* 2011;81(7):1325–33.
- [13] Gautam Suresh, Brahma Sukumar M. Detection of high impedance fault in power distribution systems using mathematical morphology. *IEEE Trans Power Syst* 2012;28(2):1226–34.
- [14] Ebron Sonja, Lubkeman David L, White Mark. A neural network approach to the detection of incipient faults on power distribution feeders. *IEEE Trans Power Delivery* 1990;5(2):905–14.
- [15] Faisal MF, Mohamed A, Shareef H. Prediction of incipient faults in underground power cables utilizing S-transform and support vector regression. *International Journal on Electrical Engineering and Informatics* 2012;4(2):186.
- [16] Dong, Xiang, Yanling Yuan, Zhongqiang Gao, Chengke Zhou, Peter Wallace, Babakalli Alkali, Bojie Sheng, and Hao Zhou. "Analysis of cable failure modes and cable joint failure detection via sheath circulating current." In 2014 IEEE Electrical Insulation Conference (EIC), pp. 294–298. IEEE, 2014.
- [17] Ghaderi Amin, Mohammadpour Hossein Ali, Ginn Herbert L, Shin Yong-June. High-impedance fault detection in the distribution network using the time-frequency-based algorithm. *IEEE Trans Power Delivery* 2014;30(3):1260–8.
- [18] Sedighzadeh M, Rezazadeh A, Elkalashy Nagy I. "Approaches in high impedance fault detection a chronological review." *Advances in Electrical and Computer Engineering* 2010;10(3):114–28.
- [19] Shi Huaitao, Guo Jin, Bai Xiaotian, Guo Lei, Liu Zhenpeng, Sun Jie. Gearbox Incipient Fault Detection Based on Deep Recursive Dynamic Principal Component Analysis. *IEEE Access* 2020;8:57646–60.
- [20] Darong Huang, Lanyan Ke, Bo Mi, Ling Zhao, Guoxi Sun. A new incipient fault diagnosis method combining improved RLS and LMD algorithm for rolling bearings with strong background noise. *IEEE Access* 2018;6:26001–10.
- [21] Ramos Maicon JS, Resener Mariana, Bretas Arturo S, Bernardon Daniel P, Leborgne Roberto C. Physics-based analytical model for high impedance fault location in distribution networks. *Electr Power Syst Res* 2020;188:106577.
- [22] Nunes, JUNg, A. S. Bretas, N. G. Bretas, A. R. Herrera-Orozco, and L. U. Iurinic. "Distribution systems high impedance fault location: A spectral domain model considering parametric error processing." *International Journal of Electrical Power & Energy Systems* 109 (2019): 227–241.
- [23] Bretas, Arturo Suman, Andrés Ricardo Herrera-Orozco, Cesar Augusto Orozco-Henao, Leonardo Ulises Iurinic, and Juan Mora-Flórez. "Incipient fault location method for distribution networks with underground shielded cables: A system identification approach." *International Transactions on Electrical Energy Systems* 27, no. 12 (2017): e2465.
- [24] Herrera-Orozco AR, Bretas AS, Orozco-Henao C, Iurinic LU, Mora-Flórez J. Incipient fault location formulation: A time-domain system model and parameter estimation approach. *Int J Electr Power Energy Syst* 2017;90:112–23.
- [25] Preston, Gary, Z. M. Radojević, Chul-Hwan Kim, and Vladimir Terzija. "New settings-free fault location algorithm based on synchronised sampling." *IET Generation, Transmission & Distribution* 5, no. 3 (2011): 376–383.
- [26] Terzija Vladimir, Radojević Zoran M, Preston Gary. Flexible synchronized measurement technology-based fault locator. *IEEE Trans Smart Grid* 2014;6(2):866–73.
- [27] Terzija Vladimir, Preston Gary, Stanojević Vladimir, Elkalashy Nagy I, Popov Marjan. Synchronized measurements-based algorithm for short transmission line fault analysis. *IEEE Trans Smart Grid* 2015;6(6):2639–48.
- [28] Habedank U. "On the mathematical description of arc behaviour in the vicinity of current zero." *et. Archiv* 1988;10:339–43.
- [29] Van der Sluis L, Rutgers WR, Koreman CGA. A physical arc model for the simulation of current zero behavior of high-voltage circuit breakers. *IEEE Trans Power Delivery* 1992;7(2):1016–22.
- [30] Schavemaker PH, Van der Sluis L. An Improved Mayr-Type Arc Model Based on Current-Zero Measurements. *IEEE Trans Power Delivery* 2000;15(2):580–4.
- [31] Yuan Ling, Sun Lin, Huaren Wu. Simulation of fault arc using conventional arc models. *Energy and Power Engineering* 2013;5(04):83.
- [32] Terzija Vladimir V, Koglin H-J. On the modeling of long arc in still air and arc resistance calculation. *IEEE Trans Power Delivery* 2004;19(3):1012–7.
- [33] Stokes AD, Oppenlander WT. Electric arcs in open air. *J Phys D Appl Phys* 1991;24(1):26.
- [34] Gammon Tammy, Matthews John. Instantaneous arcing-fault models developed for building system analysis. *IEEE Trans Ind Appl* 2001;37(1):197–203.
- [35] Gaudreau André, Koch Bohdan. Evaluation of LV and MV arc parameters. *IEEE Trans Power Delivery* 2007;23(1):487–92.
- [36] Kulkarni Saurabh, Allen Alicia J, Chopra Shivaz, Santoso Surya, Short Thomas A. Waveform characteristics of underground cable failures. In: *IEEE PES General Meeting*; 2010. p. 1–8.
- [37] Kizilcay M, Pniok T. Digital simulation of fault arcs in power systems. *Eur Trans Electr Power* 1991;1(1):55–60.
- [38] Terzija Vladimir, Preston Gary, Popov Marjan, Terzija Natasa. New static "airarc" EMTF model of long arc in free air. *IEEE Trans Power Delivery* 2011;26(3):1344–53.
- [39] Terzija VV, Koglin H-J. "Long arc in free air: laboratory testing, modelling, simulation and model-parameters estimation." *IEE Proceedings-Generation, Transmission and Distribution* 2002;149(3):319–25.
- [40] Terzija Vladimir V, Koglin H-J. A new approach to arc resistance calculation. *Electr Eng* 2001;83(4):187–92.
- [41] The Math Works, Inc., MATLAB, version 2020a (Natick, MA: The Math Works, Inc., 2020), accessed May 28, 2020, <https://www.mathworks.com/>.
- [42] Huang Zhi-Kai, Chau Kwok-Wing. A new image thresholding method based on Gaussian mixture model. *Appl Math Comput* 2008;205(2):899–907.
- [43] Moulin Pierre, Liu Juan. Analysis of multi resolution image denoising schemes using generalized Gaussian and complexity priors. *IEEE Trans Inf Theory* 1999;45(3):909–19.
- [44] Otsu Nobuyuki. A threshold selection method from gray-level histograms. *Automatica* 1975;11(285–296):23–7.
- [45] Cuevas Erik, Sosa Humberto. A comparison of nature inspired algorithms for multi-threshold image segmentation. *Expert Syst Appl* 2013;40(4):1213–9.
- [46] Guo R, Pandit SM. Automatic threshold selection based on histogram modes and a discriminant criterion. *Mach Vis Appl* 1998;10(5):331–8.